Mittuniversitetet
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Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
S - 0001 - PUO		
Kurskod / Course code: DT081A	Provkod/ Test code: T101	Tentamensdatum / Examination date: 2024-10-29
Kursnamn / Course name: Datateknik AV, Signal- och bildbehandling		
Provnamn / Test name: Tentamen		

Skriv din anonymitetskod på varje inlämnat papper Calculator: TI-82 STATS
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	4,5		16			
2	X	1,5		17			
3	X	3		18			
4	X	4		19			
5	X	6		20			
6	X	4		21			
7	X	2		22			
8	X	4		23			
9	X	5		24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		34	Betyg / Grade: B	Lärarsign./ Teachers sign			

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
16	0	ok	

2707183

Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2024-10-28 kl. 12:37:16

Example Solutions for DT081A – Signal and Image Processing Exam, 2024-10-29

1 (1 + 1 + 1 + 1 + 1 = 5 p)

a)
The process of converting a continuous-time signal into a discrete-time signal by taking periodic samples, $x[n] = x(nT_s)$ where T_s is the sampling interval.

b)
The process of mapping a continuous range of values to a finite set of discrete levels, often in analog-to-digital conversion.

c)
The output of a system when an impulse signal is applied at its input. It characterizes the system's behavior.

d)
The steady-state response of a system to sinusoidal inputs at varying frequencies.

e)
A technique used to reduce spectral leakage in signal analysis by applying a window function to a signal before performing a Fourier transform.

2 (3 + 2 = 5 p)

a)
$y[n] = s[n] * h[n]$

$y[0] = (3)(-0.5) = -1.5$
$y[1] = (7)(-0.5) + (3)(0) = -3.5$
$y[2] = (10)(-0.5) + (7)(0) + (3)(0.5) = -5 + 1.5 = -3.5$
$y[3] = (3)(-0.5) + (10)(0) + (7)(0.5) = -1.5 + 3.5 = 2$
...

$y[n] = \{-1.5, -3.5, -3.5, 2, 7.5, 1.5, -2.5\}$

b) This filter approximates a differentiation operation, emphasizing changes in the signal, which is characteristic of a high-pass filter.

3 (1 + 2 = 3 p)

a) The system diagram shows both a feed-forward and feedback branch. The latter implies that the system will produce an infinite impulse response IIR.

b)
$y(n) = x(n) + 2x(n-1) + x(n-3) - 3y(n-1) + 2y(n-3)$

4 (2 + 3 = 5 p)

a)
Extend spectrum plot to at least $-2\pi < \Omega < 2\pi$, where the continous spectrum is mirrored on the y-axis and replicated over the whole Ω axis, with a periodicity of 2π.

b)
$f_s \geq 2f_{max}$
The lowest sampling frequency depends on the signal bandwidth. The figure shows that $f_{max} = \frac{3}{4}\pi\frac{f_s}{2\pi} = \frac{3}{8}f_s = \frac{150}{8} = 18.75Hz$.
Hence $f_s \geq 37.5Hz$.

5 (2 + 4 = 6 p)

a) Amplitude and phase relationship:
$|Y(f)| = |H(f)||X(f)|$
$\angle Y(f) = \angle H(f) + \angle X(f)$

b) Sketch spectra

$$|Y(f)| = \begin{cases} 8f & \text{for } 0 \leq f \leq 10 \\ 80 - 4(f-10) & \text{for } 10 < f \leq 30 \\ 0 & \text{for } f < 0 \text{ or } f > 30 \end{cases}$$

and

$arg(Y(f)) = 0 \forall f$

6 (2 + 2 = 4 p)

a) Window iii) has the narrowest main lobe, which provides the best frequency separation between closely spaced frequency components allowing for better distinction.
b) Window iv) has zero outside of main lobe, which effectively eliminates spectral leakage in the frequency domain ensuring that frequency components are not contaminated by artifacts from distant frequencies.

7 (3 p)

At low frequencies the system has an amplification of 1, whereas higher frequencies are amplified more. Combined this results in high frequency amplification (not to be confused with a high pass filter), which if applied to an image would enhance edges.

8 (1 + 1 + 1 + 1 + 1 = 5 p)

- $1 \rightarrow ii$
  Only the center element is used, and reduce the input with 0.5.
- $2 \rightarrow iv$
  A vertical filter emphasized vertical edges only.
- $3 \rightarrow v$
  Edge detecting kernel with elements summing to 0, only shows edges.
- $4 \rightarrow iii$
  Avering kernel smooths the image as a low-pass filter.
- $5 \rightarrow i$
  Edge detecting kernel summed with a unit kernel translates kernel to one with center element = 5, with elements summing to 1.

9 (6 p)

$x(i,j)$ has several non-zero values, that each contribute to the $y(i,j)$ through convolution with $h(i,j)$. From examining the isolated scaled impulse responses we can determine that $h(i,j)$ has a cross-shaped pattern. By subtracting one scaled impulse reponse from another we can identify the impulse response as.

$$h(i,j) = \begin{bmatrix} 0 & 1/8 & 0 \\ 1/8 & 1/2 & 1/8 \\ 0 & 1/8 & 0 \end{bmatrix}$$

Lärarens anteckning / Teachers note:

1.

a) Sampling - Sampling is a concept when you sample/select certain points of a signal in a selective manner in order to represent the signal with fewer points. Could for an example be during a conversion from a continuous signal to a discrete signal.

b) Quantization - Quantization is a concept where you change the range of possible values a signal could obtain. It could for an example be going from 8-bit representation for the pixels in an image to 4-bits.

c) Impulse response - Impulse response describes how a system response to an input signal in the time domain. By knowing the impulse response the resulting output of an input signal could be calculated.

→

Lärarens anteckning / Teachers note:

1. →

d) Frequency response - The frequency response describes how a system responds to an input signal in the frequency domain. It could for an example describe how different frequency components are amplified or phase shifted in the output.

e) Windowing - When processing signals it is often desired to process them in a shorter time-frame than it is defined. Therefore the signals are truncated in a desired way which is called windowing. For this, windowing functions are used such as rectangular, Hann and Hamming windows.

1

1

4,5

Lärarens anteckning / Teachers note:

2.

a) $s(n) = \{3, 7, 10, 3, -5\}$

$h(n) = \{-0,5 \;,\; 0 \;,\; 0,5\}$

The resulting sequence $y(n)$ is described by the convolution:

$$y(n) = \sum_{k=0}^{n} s(k)\, h(n-k)$$

Calculating from $n = 0, 1, 2, 3, 4$

Assuming that the resulting sequence is the same length and $h(n) = 0$, $n > 2$

For $n=0$

$$y(0) = \sum_{k=0}^{0} s(0)\, h(0-0) = 3 \cdot (-0,5) = -1,5$$

For $n=1$

$$y(1) = \sum_{k=0}^{1} s(k)\, h(1-k) = s(0)\, h(1) + s(1)\, h(0)$$

$$= 3 \cdot 0 + 7 \cdot (-0,5) = 3,5$$

For $n=2$

$$y(2) = \sum_{k=0}^{2} s(k)\, h(2-k) = s(0)\, h(2) + s(1)\, h(1)$$

$$+ s(2)\, h(0) = 3 \cdot 0,5 + 7 \cdot 0 + 10 \cdot (-0,5) =$$

$$= -3,5$$

→

Lärarens anteckning / Teachers note:

2. → a)

For $n=3$

$$y(3) = \sum_{k=0}^{3} s(k)\,h(3-k) = s(0)\,h(3)$$

$$+ s(1)\,h(2) + s(2)\,h(1) + s(3)\,h(0) =$$

$$= 3\cdot 0 + 7\cdot 0{,}5 + 10\cdot 0 + 3\cdot(-0{,}5) =$$

$$= 2$$

For $n=4$

$$y(4) = \sum_{k=0}^{4} s(k)\,h(4-k) = s(0)\,h(4)$$

$$+ s(1)\,h(3) + s(2)\,h(2) + s(3)\,h(1)$$

$$+ s(4)\,h(0) = 3\cdot 0 + 7\cdot 0 + 10\cdot 0{,}5 +$$

$3\cdot 0 + (-5)\cdot(-0{,}5) = 7{,}5$

The resulting sequence is

$y(n) = \{y(0), y(1), y(2), y(3), y(4)\} =$

$y(n) = \{-1.5,\ 3.5,\ -3.5,\ 2,\ 7{,}5\}$

1,5

2. b) The operation that this filter approximates is compacting the range of values in the resulting sequence by minimizing the average and the spread of the sequence. This could also be seen as a smoothing operation.

Lärarens anteckning / Teachers note:

0

1,5

Lärarens anteckning / Teachers note:

3. a)

This system is an infinite impulse respons (IIR) system. The reason for this is because it contains two feedback terms: $-3y(n-1)$, $2y(n-3)$. This means that the output will be affected by previous output resulting in that the system will have an infintie impulse response. 1

b) The differential equation of the system is:

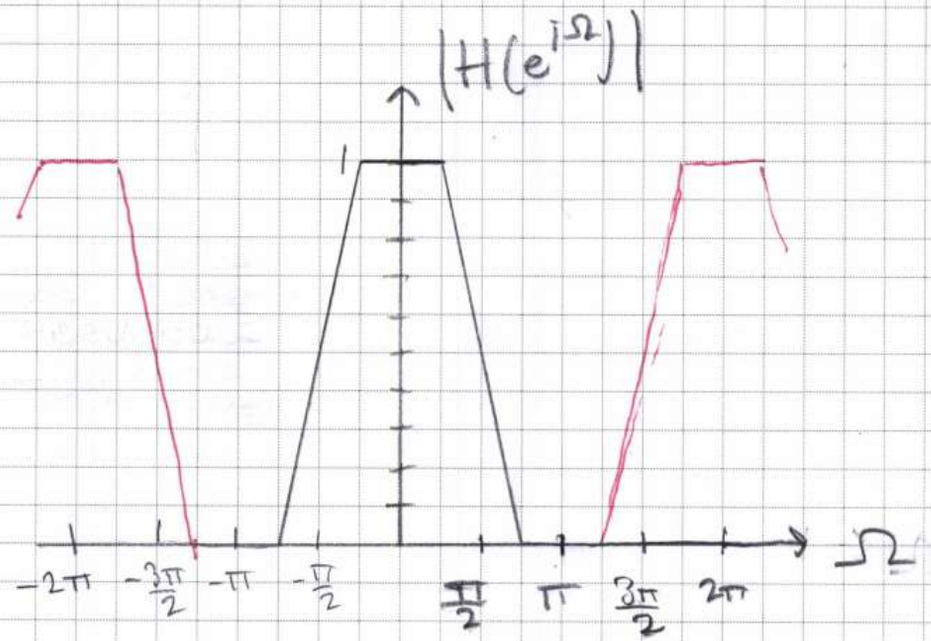
$$y(n) = x(n) + 2x(n-1) + x(n-3) - 3y(n-1) + 2y(n-3)$$ 2

The left part of the systems contains 3 forward terms: $x(n)$, $2x(n-1)$ $x(n-3)$. The right part has two feedback terms: $-3y(n-1)$, $2y(n-3)$

/3

Lärarens anteckning / Teachers note:

4. a) $f_s = 50$ kHz



1

b) Since the frequency range Ω has been normalized, (aaming with f_s) then:

$\Omega = \frac{w}{f_s}$. According to Nyquist sampling theorem, the lowest possible sampling frequency must be atleast twice the highest frequency in the signal.

$f_s \geq 2f_{max}$. By the amplitude spectrum, the highest frequency component is $\Omega_{max} = \frac{3\pi}{4}$ since the amplitude is zero for frequency components higher then that.

Skriv ej på baksidan! Do not write on the other side!

Lärarens anteckning / Teachers note:

→ 4. b) To then calculate the highest frequency component in order to know the minimum f_s:

$$\Omega_{max} = \frac{\omega_{max}}{f_s} \Rightarrow \omega_{max} = \frac{3\pi}{4} \cdot 50 \cdot 10^3$$

$$\omega_{max} = 2\pi f_{max} \Rightarrow \frac{3\pi}{4} \cdot 50 \cdot 10^3 = 2\pi f_{max} \Rightarrow$$

$$f_{max} = \frac{3\pi \cdot 50 \cdot 10^3}{4 \cdot 2\pi} = 18750$$

The maximum frequency is then:

$f_{max} = 18{,}75$ kHz.

Then the lowest sampling frequency possible in order to avoid aliasing and reconstruct it exactly is according to Nyquist theorem:

$$f_s \geq 2f_{max}$$

$$f_s = 2f_{max} \quad \text{(lowest)}$$

$$f_s = 2 \cdot 18{,}75 \text{ kHz} = 37{,}5 \text{ kHz}$$

3/4

Lärarens anteckning / Teachers note:

5. a) The relationship between the input and output is:

$$|Y(f)|e^{j\arg Y(f)} = |H(f)||X(f)|e^{j(\arg H(f) + \arg X(f))}$$

since $Y(f) = H(f)X(f)$.

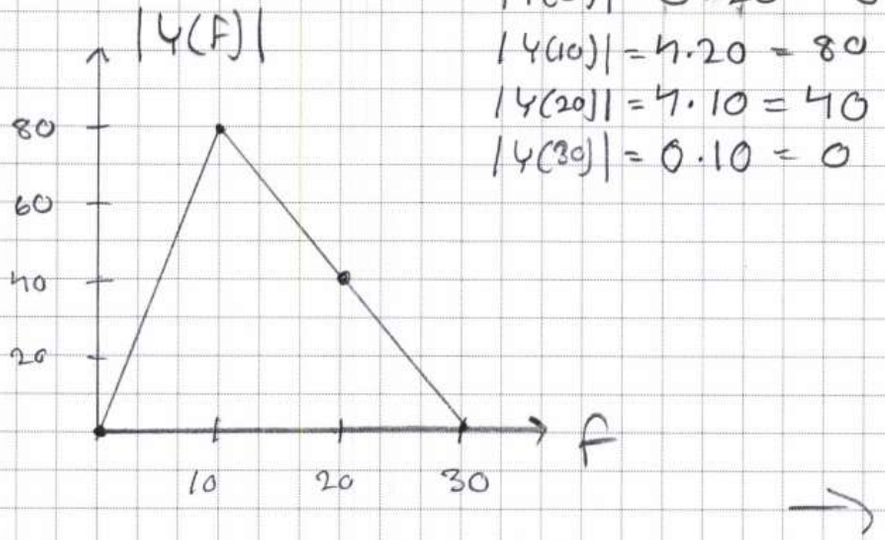
The relationship between amplitudes and phases of the input and output is then:

Amplitude: $|Y(f)| = |H(f)||X(f)|$

Phase: $\arg Y(f) = \arg H(f) + \arg X(f)$ 2

b) Using $|Y(f)| = |H(f)||X(f)|$ to draw the amplitude spectrum for $y(t)$:

$|Y(0)| = 0 \cdot 20 = 0$
$|Y(10)| = 4 \cdot 20 = 80$
$|Y(20)| = 4 \cdot 10 = 40$
$|Y(30)| = 0 \cdot 10 = 0$

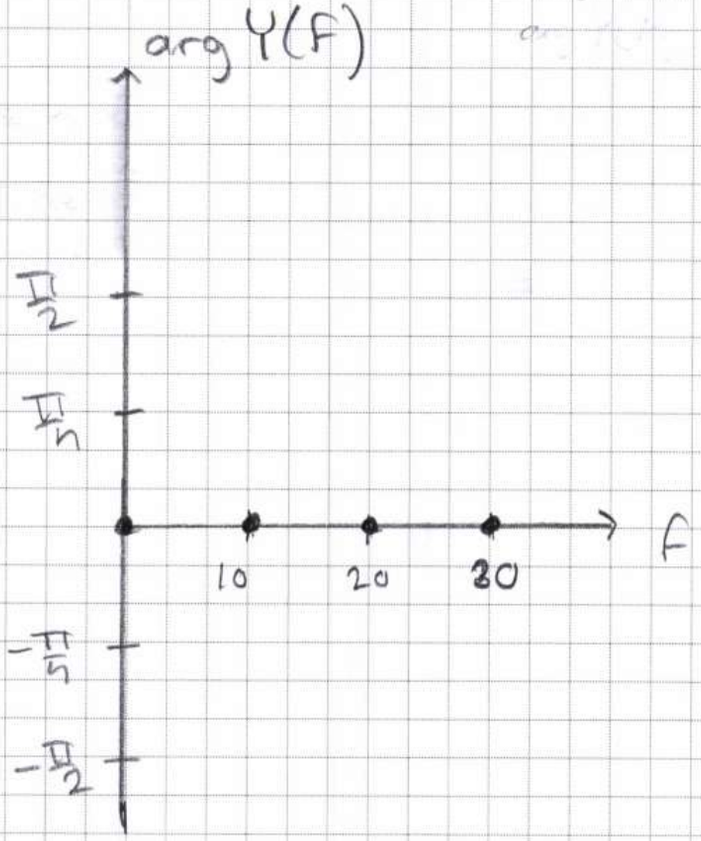


→

Lärarens anteckning / Teachers note:

→ 5. b)

Using $\arg Y(f) = \arg H(f) + \arg X(f)$ to draw the phase spectrum:



$\arg Y(0) = 0 + 0 = 0$

$\arg Y(10) = \frac{\pi}{4} + (-\frac{\pi}{4}) = 0$

$\arg Y(20) = \frac{\pi}{4} + (-\frac{\pi}{4}) = 0$

$\arg Y(30) = \frac{\pi}{2} + (-\frac{\pi}{2}) = 0$

4

/6

Lärarens anteckning / Teachers note:

6. a) (Assuming that damping the sidebands are of less interest)
The window function that achieves the best frequency separation is window iii), The reason for this is because it has the narrowest main lobe which makes it most suitable for separating frequencies close to each other.

2

b) The best window to minimize spectral leakage is window iv). The reason for this is because it has the best damping of the sidelobes (zero), which minimizes the frequency component around the main frequency infinitely.

2

4

Lärarens anteckning / Teachers note:

7. Based on the figure, it could be seen that the 2D system contains more higher frequency content than lower frequency content. The reason for this is because the magnitude are higher farther away from the middle which is where the higher frequency components are located. At the edges the magnitude is the highest which indicates on that the highest frequency components are most present. In the middle where the lower frequencies are found, the magnitude is low, indicating on fewer low frequency components present.

Using signal rather than system nomenclature

2

Lärarens anteckning / Teachers note:

8. Kernel 1: This kernel halfs the intensity of each pixel and leaves the surrounding unchanged. This would result in that each pixel has half of its original intensity in the resulting image, resulting in a darkened image. This kernel corresponds to result ii) 1

Kernel 2: This kernel will enhance vertical edges, and suppress horizontal ones. This will result in a dark image where the vertical edges remains. Therefore this kernel corresponds to result v) 0,5

Kernel 3: This kernel will also result in a sharpened image similar to kernel 5. It will increase the intensity of each pixel while decreasing surrounding ones. This kernel therefore corresponds to result i) 0,5

Lärarens anteckning / Teachers note:

8.

Kernel 4: This kernel will result in a smoothen/blurred image, since it takes $\frac{1}{9}$ of each pixel value and spreads it out to neighbouring pixels. Therefore it corresponds to result iii) 1

Kernel 5: This kernel will sharpen the image. Since it will increase the intensity of each pixel by 5 and reduce the surrounding the filtered image will look sharper. Therefore this kernel corresponds to result i) 1

4

Lärarens anteckning / Teachers note:

9. By examining the values in $x(i,j)$ which are not 0, this hints about the middle value of the PSF. Since transform from $x(i,j) \rightarrow y(i,j)$ could be seen as stamping the PSF on each value, the middle value of the PSF is then:

$x(1,1) = 112 \rightarrow y(1,1) = 56$

$x(3,0) = 80 \rightarrow y(3,0) = 40$

$x(3,2) = 64 \rightarrow y(3,2) = 32.$

⋮ ⋮

The middle of the PSF $h(i,j)$ is then: $h(1,1) = \frac{1}{2}$ since the non-zero values in $x(i,j)$ is halfed in $y(i,j)$.

Looking at the relationship of the values at the adjacent positions of the non-zero values in $x(i,j)$:

Values surrounding $y(1,1) = 56$ in $y(i,j)$ are equal to 14.

→

Lärarens anteckning / Teachers note:

9. → The values surrounding $y(3,0) = 40$ not affected by other non-zero values in $x(i,j)$ has values of 10.

These values are both $\frac{1}{8}$ of the non-zero values in $x(i,j)$

$x(1,1) = 112 \quad 112 \cdot \frac{1}{8} = 14$

$x(3,0) = 80 \quad 80 \cdot \frac{1}{8} = 10$

The adjacent values of the middle in the PSF ($h(1,1)$) will then have values of $\frac{1}{8}$.

The PSF will not have values in its corners which could be verified by $x(1,1)$ since it has not spread to $y(0,0)$ and $y(0,2)$.

The PSF, $h(i,j)$ is therefore:

$$h(i,j) = \begin{bmatrix} 0 & \frac{1}{8} & 0 \\ \frac{1}{8} & \frac{1}{2} & \frac{1}{8} \\ 0 & \frac{1}{8} & 0 \end{bmatrix}.$$

The PSF will half the current value and spread $\frac{1}{8}$ of it to the pixels near it.

5